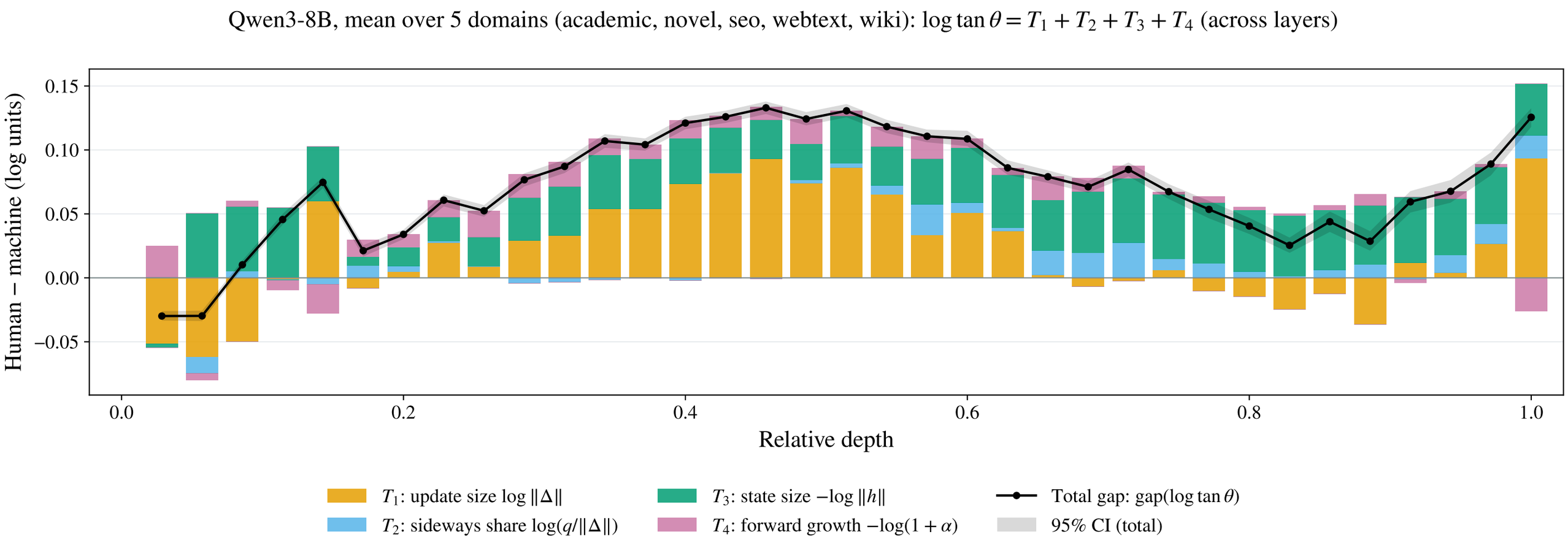
Qwen3-8B, mean over 5 domains (academic, novel, seo, webtext, wiki): log tan θ = T1 + T2 + T3 + T4 (across layers)
Human − machine (log units)
0.15
0.10
0.05
0.00
−0.05
0.0
0.2
0.4
0.6
0.8
1.0
Relative depth
T1: update size log ‖Δ‖
T2: sideways share log(q/‖Δ‖)
T3: state size −log ‖h‖
T4: forward growth −log(1 + α)
Total gap: gap(log tan θ)
95% CI (total)